

BLACK FOREST HACKATHON 2026

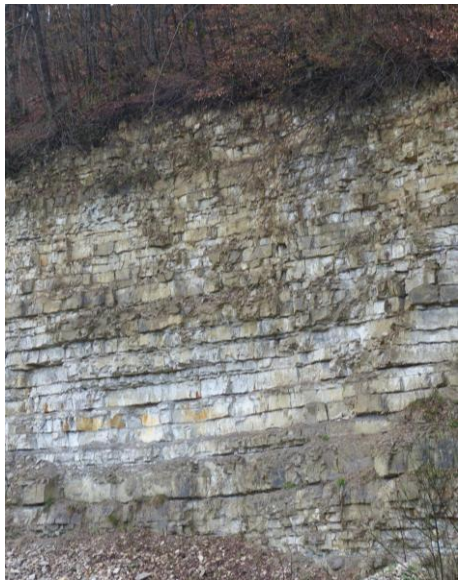
READING THE GROUND

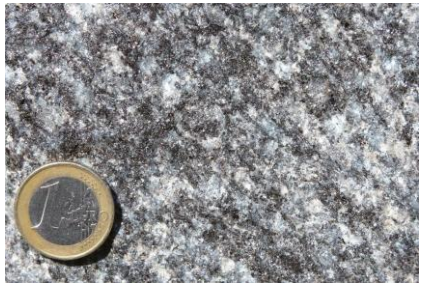
Jan Derks, Tobias Braun, Gabriel Lehmann, Charlotte Kiesele

Why it matters

EVERY PROJECT STARTS WITH UNDERSTANDING THE GROUND

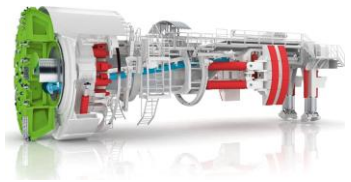
GEOLOGY

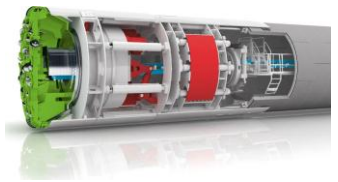


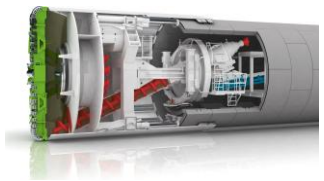




MACHINE







Massive
Shields and cutterheads built at the scale of buildings

High-tech
Drives, support pressure and sensors tuned to the rock

Worldwide
Each machine tailored to its project and its geology

No two tunnels cross the same ground – no two machines are the same.

Our window into the subsurface

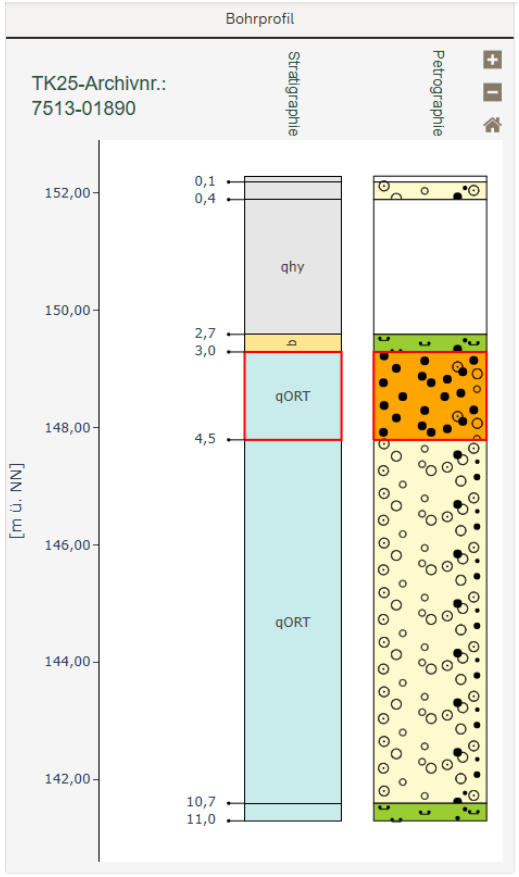
THE BOREHOLE TELLS THE STORY

01 **Elevation**
Absolute height of the borehole start

02 **Depth below surface**
Where each layer begins and ends

03 **Age & color code**
Formation abbreviations and their colors

04 **Rocktype, pattern & colour**
What the rock or soil actually is





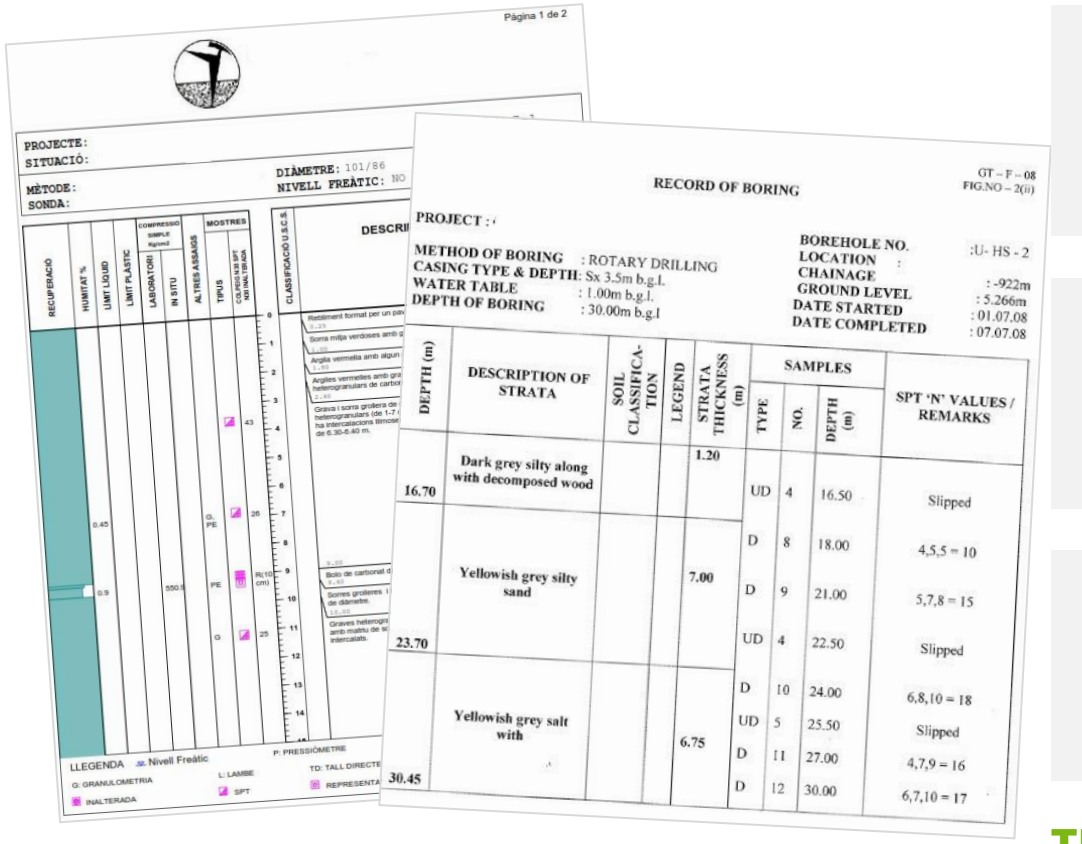


Boreholes are, most of the time, the only information that describes the subsurface accurately.

Every metre of a log shapes how a tunnel boring machine is designed.

The problem

KNOWLEDGE LOCKED IN PAPER



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PROJECTE:
SITUACIÓ:
MÈTODE:
SONDA:
DIÀMETRE: 101/86
NIVELL FREÀTIC: NO

RECUPERACIÓ | HUMITAT % | LIMIT LÍQUID | LIMIT PLÀSTIC | COMPRESSIÓ SIMPLE Kp/cm2 | LABORATORI | IN SITU | ALTRES ASSAIGS | MOSTRES TIPUS | CLASSIFICACIÓ U.S.C.S. | DESCRI

LLEGENDA Nivell Freàtic
G: GRANULOMETRIA
INALTERADA
L: LAMBE
SPT
P: PRESSIÒMETRE
TD: TALL DIRECTE
REPRESENTA

RECORD OF BORING

GT – F – 08
FIG.NO – 2(ii)

PROJECT :

METHOD OF BORING : ROTARY DRILLING
CASING TYPE & DEPTH: Sx 3.5m b.g.l.
WATER TABLE : 1.00m b.g.l.
DEPTH OF BORING : 30.00m b.g.l

BOREHOLE NO. :U- HS - 2
LOCATION :
CHAINAGE : -922m
GROUND LEVEL : 5.266m
DATE STARTED : 01.07.08
DATE COMPLETED : 07.07.08

DEPTH (m)	DESCRIPTION OF STRATA	SOIL CLASSIFICATION	LEGEND	STRATA THICKNESS (m)	SAMPLES TYPE	SAMPLES NO.	SAMPLES DEPTH (m)	SPT 'N' VALUES / REMARKS
	Dark grey silty along with decomposed wood			1.20	UD	4	16.50	Slipped
16.70					D	8	18.00	4,5,5 = 10
	Yellowish grey silty sand			7.00	D	9	21.00	5,7,8 = 15
23.70					UD	4	22.50	Slipped
	Yellowish grey salt with			6.75	D	10	24.00	6,8,10 = 18
					UD	5	25.50	Slipped
					D	11	27.00	4,7,9 = 16
30.45					D	12	30.00	6,7,10 = 17

01 Unstructured
Free-form layouts, no common schema

02 Multilingual
English, French, Spanish and more

03 Local geology terms
Regional names and abbreviations

04 Variable formats
Every company and decade has its own log

05 Mixed quality
From scanned paper to native digital

06 Color vs. B/W
Patterns and colors carry meaning

The knowledge exists – but no machine can read it yet.

The hackathon challenge

LET'S TEACH MACHINES TO READ THE GROUND

01

OCR
Read scans and PDFs

02

Metadata
Location, date, elevation

03

Extract
Depth, rocks (lithology), age (stratigraphy)

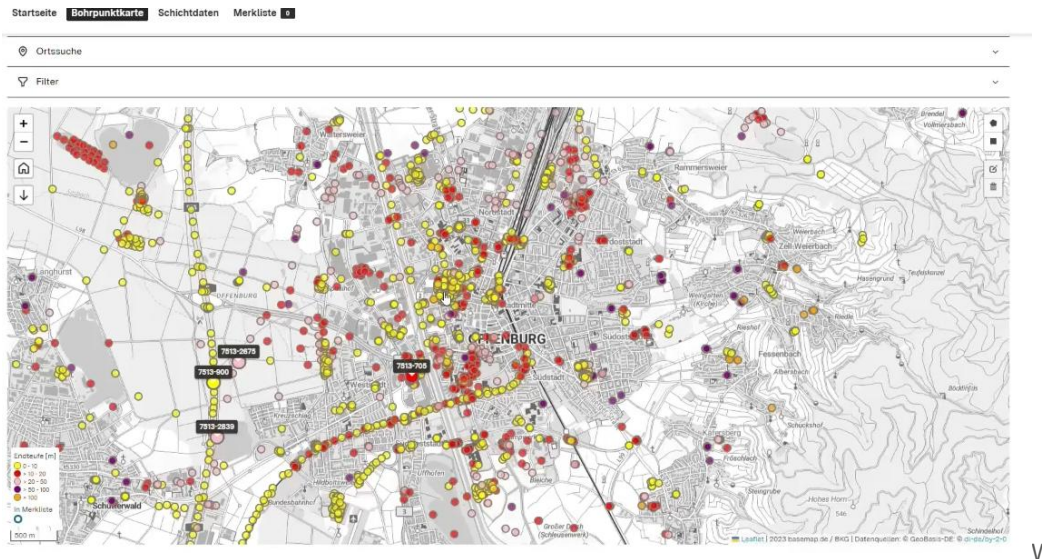
04

Map & profile
Visualize boreholes in context

05

Search
Find the right log in seconds

DATA SECURITY ACROSS EVERY STEP



From unstructured logs to geotechnical parameters

A structured, searchable borehole base that feeds geology straight into TBM design – faster, more consistent, for projects worldwide.

Join us. Turn paper into tunnels.

We are here…Example: bohrungen.lgrb-bw.de – borehole map, Baden-Württemberg